KS3 DIGITAL TECHNOLOGY

ENCRYPTION AND CRYPTOGRAPHY

From ancient ciphers to the padlock in your browser: how encryption keeps secrets, how codes are broken, how strangers share keys safely, how passwords are stored, and why quantum computers are changing the rules.

LESSONS

1 Breaking codes by hand

2 Keys and brute force

3 Sharing secrets with strangers

4 The padlock in your browser

5 Hashes and private chats

6 Quantum threats and the encryption debate

LESSON 1 OF 6

BREAKING CODES BY HAND

A substitution cipher swaps each letter for another.

ENCRYPTION AND CRYPTOGRAPHY

STARTER · 5 MINUTES

DO NOW

Decode this Caesar cipher (shift 3): FRGH EUHDNHU. What's the weakness of this cipher?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Encrypt and decrypt with substitution ciphers and break them using frequency analysis

SUCCESS CRITERIA: I CAN...

- ✓ I can explain plaintext, ciphertext and key
- ✓ I can encrypt with a substitution cipher
- ✓ I can use frequency analysis to break a cipher

THE LANGUAGE OF SECRETS

- Cryptography: the science of keeping information secret
- Plaintext: the readable message
- Ciphertext: the scrambled message
- Key: the secret that locks and unlocks it

KEY WORDS

CRYPTOGRAPHY	The science of keeping information secret, for example by encryption.
PLAINTEXT	The original, readable message before encryption.
CIPHERTEXT	The scrambled message produced by encryption.
KEY	The secret value used to encrypt and decrypt a message.
FREQUENCY ANALYSIS	Breaking a cipher by counting how often letters appear and matching them to the most common letters in the language.

SUBSTITUTION CIPHERS

- Each letter is replaced by a different letter, in any order
- 26! (about 4×10^{26}) possible keys: far too many to try them all by hand
- Used for centuries by spies and royalty
- So how could anyone break it?

FREQUENCY ANALYSIS

- In English, E, T and A are the most common letters
- Count the letters in the ciphertext: the most common probably stands for E
- Look for patterns: one-letter words are probably A or I; "the" is very common
- First described by the Arab scholar Al-Kindi over 1,100 years ago

ACTIVITY · 25 MINUTES · PAIRS

CRACK THE SPY MESSAGE

1. Your teacher gives you a 200-letter message encrypted with a substitution cipher and an English letter-frequency chart.

2. Count how many times each letter appears in the ciphertext and record the top five.

3. Match the most common letters to E, T and A. Fill them in and look for word patterns.

4. Keep going until you can read the message.

In your workbook, answer:

? Which three ciphertext letters were most common, and what did they stand for?

? Explain why frequency analysis works on substitution ciphers.

Your workbook: Lesson 1 · tick each step off as you go

QUICK CHECK

1. What is ciphertext?

A The scrambled, encrypted message
B The original readable message
C A type of font
D The secret key

2. What is the key in cryptography?

A A keyboard button
B The plaintext
C The secret used to encrypt and decrypt
D A password manager

3. Which is the most common letter in English?

A E
B K
C Z
D Q

4. Why is a Caesar cipher easy to break?

A It's always in capitals
B It needs a computer
C It uses numbers
D There are only 25 possible shifts to try

QUICK CHECK: ANSWERS

1. **A: The scrambled, encrypted message**
   Ciphertext is encrypted text.

2. **C: The secret used to encrypt and decrypt**
   The key locks and unlocks.

3. **A: E**
   E makes up about 13% of letters.

4. **D: There are only 25 possible shifts to try**
   Brute force is quick.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between plaintext and ciphertext.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain plaintext, ciphertext and key
- ✓ I can encrypt with a substitution cipher
- ✓ I can use frequency analysis to break a cipher

LESSON 2 OF 6

KEYS AND BRUTE FORCE

Modern encryption uses mathematics and very long binary keys.

STARTER · 5 MINUTES

DO NOW

A bike lock has 3 dials of 0-9. How many combinations? What if it had 4 dials?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain why key length matters and describe symmetric encryption

SUCCESS CRITERIA: I CAN...

- ✓ I can explain a brute-force attack
- ✓ I can explain why longer keys are stronger
- ✓ I can describe symmetric encryption

TRY EVERY KEY

- A brute-force attack tries every possible key until one works
- 3-dial lock: 1,000 combinations; 4 dials: 10,000
- Computers can try billions of keys a second
- Defence: make the number of keys astronomically large

KEY WORDS

BRUTE-FORCE ATTACK	Trying every possible key or password until the right one is found.
KEY LENGTH	The number of bits in an encryption key; each extra bit doubles the number of possible keys.
SYMMETRIC ENCRYPTION	Encryption where the same key is used to encrypt and decrypt.
AES	Advanced Encryption Standard: the world's most widely used symmetric encryption method.

EVERY BIT DOUBLES IT

- Keys are binary numbers: each bit is 0 or 1
- 8-bit key: 256 possible keys; 128-bit key: about 3.4×10^{38}
- Each extra bit doubles the number of keys
- Brute-forcing a 128-bit key would take longer than the age of the universe

ACTIVITY · 20 MINUTES · PAIRS

BRUTE-FORCE RACE

1. Partner A sets a secret 2-digit code (00-99); Partner B tries codes one at a time and counts the guesses. Swap.

2. Now try a 3-digit code. How long do you think it would take?

3. Use the calculator to work out how long a computer trying 1 billion keys per second would take to try every 56-bit and every 128-bit key.

In your workbook, answer:

? How many guesses did it take for 2 digits? How many possible codes were there for 3 digits?

? Explain why adding bits to a key makes it so much stronger.

Your workbook: Lesson 2 · tick each step off as you go

SYMMETRIC ENCRYPTION

- The same key encrypts and decrypts
- Fast, so it's used for most data: Wi-Fi, files, video calls
- AES (Advanced Encryption Standard) is the world standard
- Problem: how do you share the key safely?

QUICK CHECK

1. What is a brute-force attack?

A Using a virus
B Guessing the password once
C Smashing the computer
D Trying every possible key until one works

2. How many keys does a 4-bit key have?

A 32
B 16
C 4
D 8

3. What happens to the number of possible keys when you add one bit?

A It goes up by one
B It halves
C It doubles
D It stays the same

4. What is symmetric encryption?

A Using the same key to encrypt and decrypt
B Encryption that looks symmetrical
C Using two different keys
D Encryption without a key

QUICK CHECK: ANSWERS

1. D: Trying every possible key until one works

It tries all keys.

2. B: 16

$2^4 = 16$.

3. C: It doubles

Each bit doubles the possibilities.

4. A: Using the same key to encrypt and decrypt

One shared key.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why a 128-bit key is much more secure than a 40-bit key.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain a brute-force attack
- ✓ I can explain why longer keys are stronger
- ✓ I can describe symmetric encryption

LESSON 3 OF 6

SHARING SECRETS WITH STRANGERS

Public-key (asymmetric) cryptography uses a pair of keys: a public key anyone can use to lock a message and a private key only the owner can use to unlock it.

STARTER · 5 MINUTES

DO NOW

You want to send a locked box to someone you've never met, but you can't send the key in the post. How could you do it?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how public-key cryptography lets people communicate securely without first sharing a secret key

SUCCESS CRITERIA: I CAN...

- ✓ I can explain the key-sharing problem
- ✓ I can describe how public and private keys work
- ✓ I can explain the idea of a one-way function

THE PADLOCK TRICK

- You send me an open padlock (your public key) and keep its key (your private key)
- I put my message in a box and snap your padlock shut
- Anyone can see the padlock, but only you can open it
- No secret ever had to travel through the post

KEY WORDS

PUBLIC-KEY CRYPTOGRAPHY	Encryption using a pair of linked keys: a public key to encrypt and a private key to decrypt.
PUBLIC KEY	The key that is shared openly and used to encrypt messages to its owner.
PRIVATE KEY	The key that is kept secret and used to decrypt messages encrypted with the matching public key.
ONE-WAY FUNCTION	A calculation that is easy to do but extremely hard to reverse.

ACTIVITY · 20 MINUTES · GROUPS OF 3

THE PADLOCK POST OFFICE

1. Act it out: Alice, Bob and Eve (the eavesdropper). Your teacher gives you a box, padlocks and keys (or paper versions).

2. Alice must send Bob a secret note. Eve sees everything that passes between them. Use only padlocks: no keys may be sent!

3. Then try the "double padlock" method: Alice locks, Bob adds his lock, Alice removes hers, Bob removes his.

4. Record each step and check Eve never had a chance to open the box.

In your workbook, answer:

? Describe how Alice sent the secret without ever sending a key.

? How does this model public-key cryptography?

Your workbook: Lesson 3 · tick each step off as you go

PUBLIC AND PRIVATE KEYS

- Public key: shared with everyone, used to encrypt messages to you
- Private key: kept secret, used to decrypt them
- The two keys are mathematically linked
- Also called asymmetric encryption

EASY ONE WAY, HARD TO REVERSE

- Multiplying two big prime numbers is easy
- Finding which two primes made a huge number is incredibly hard
- RSA encryption relies on this one-way problem
- 13 × 17 = 221 is easy; what two primes make 6,557?

QUICK CHECK

1. What problem does public-key cryptography solve?

A Fixing broken passwords
B Sharing a key safely with someone you've never met
C Making computers faster
D Storing more data

2. Which key do you share with everyone?

A Your public key
B Your private key
C Both keys
D Neither key

3. A message encrypted with Ben's public key can be decrypted by…

A Ben's private key only
B anyone with Ben's public key
C anyone at all
D nobody

4. Why is RSA hard to break?

A It changes every second
B It's written in Latin
C It uses passwords
D Finding the two large primes that make a huge number is extremely hard

QUICK CHECK: ANSWERS

1. **B: Sharing a key safely with someone you've never met**
   It solves key distribution.

2. **A: Your public key**
   Public keys are meant to be shared.

3. **A: Ben's private key only**
   Only the private key decrypts.

4. **D: Finding the two large primes that make a huge number is extremely hard**
   It relies on factoring being hard.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how public and private keys are used to send a secret message to Sam.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain the key-sharing problem
- ✓ I can describe how public and private keys work
- ✓ I can explain the idea of a one-way function

LESSON 4 OF 6

THE PADLOCK IN YOUR BROWSER

HTTPS protects web traffic using TLS.

STARTER · 5 MINUTES

DO NOW

You're on public Wi-Fi in a café. Who could see the websites you visit and what you type?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how HTTPS uses encryption and certificates to protect communication on the web

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what HTTPS protects against
- ✓ I can describe what a digital certificate does
- ✓ I can outline the steps of an HTTPS connection

HTTP VS HTTPS

- HTTP sends web pages as plain text: anyone on the network could read them
- HTTPS = HTTP Secure: everything is encrypted
- Protects passwords, card numbers and messages from eavesdroppers
- Look for https:// and the padlock (or settings icon) in the address bar

KEY WORDS

HTTPS	Hypertext Transfer Protocol Secure: the encrypted version of the protocol used to load web pages.
TLS	Transport Layer Security: the protocol that encrypts HTTPS connections.
DIGITAL CERTIFICATE	An electronic document, issued by a trusted authority, that proves a website's identity.
SESSION KEY	A temporary secret key agreed for one connection and used for symmetric encryption.

WHO ARE YOU REALLY?

- A digital certificate proves a website is who it says it is
- Issued by a trusted certificate authority (CA)
- Your browser checks it automatically
- Warning screens mean something is wrong: don't continue

THE TLS HANDSHAKE (SIMPLIFIED)

- 1 Browser says hello; website sends its certificate and public key
- 2 Browser checks the certificate
- 3 They use public-key cryptography to agree a secret session key
- 4 Everything after that is encrypted with fast symmetric encryption (like AES)

ACTIVITY · 20 MINUTES · GROUPS OF 3

HANDSHAKE ROLE PLAY

1. Take the roles of Browser, Website and Eavesdropper. Use the role cards your teacher gives you.

2. Act out the five handshake steps, passing "messages" on cards. The Eavesdropper copies every card they see.

3. At the end, check: what did the Eavesdropper learn? Could they read the final message?

In your workbook, answer:

? List what the Eavesdropper saw and explain why they still couldn't read the message.

? A browser shows "Your connection is not private". What should you do and why?

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What does HTTPS do?

A Blocks all adverts
B Encrypts data between your browser and a website
C Deletes cookies
D Makes websites load faster

2. What does a digital certificate prove?

A That a site has no adverts
B That a website is who it claims to be
C That a website is honest
D That the site is free

3. Does a padlock in the address bar mean a website is safe to trust?

A Only on phones
B No: it means the connection is encrypted, but scam sites can have HTTPS too
C Only for shops
D Yes, always

4. Why does HTTPS use symmetric encryption for the main data?

A Public-key encryption doesn't work
B It needs no key
C It's much faster
D It's the only kind

QUICK CHECK: ANSWERS

1. **B: Encrypts data between your browser and a website**
HTTPS secures the connection.

2. **B: That a website is who it claims to be**
Certificates confirm identity.

3. **B: No: it means the connection is encrypted, but scam sites can have HTTPS too**
Encryption isn't the same as honesty.

4. **C: It's much faster**
Symmetric encryption is efficient.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why logging in to a website over HTTP on public Wi-Fi is risky.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what HTTPS protects against
- ✓ I can describe what a digital certificate does
- ✓ I can outline the steps of an HTTPS connection

LESSON 5 OF 6

HASHES AND PRIVATE CHATS

Websites shouldn't store your actual password.

STARTER · 5 MINUTES

DO NOW

If a website emails you your old password when you click "forgot password", what does that tell you about how it stores passwords?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how hashing protects stored passwords and how end-to-end encryption protects messages

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what a hash function does
- ✓ I can explain why passwords are hashed and salted
- ✓ I can describe end-to-end encryption

FINGERPRINTS FOR DATA

- A hash function turns any input into a fixed-length "fingerprint"
- Same input = same hash; tiny change = totally different hash
- One-way: you can't turn a hash back into the input
- Used for passwords, checking downloads and digital signatures

KEY WORDS

HASH	A fixed-length "fingerprint" made from data by a one-way function.
SALT	Random data added to a password before hashing so that identical passwords get different hashes.
END-TO-END ENCRYPTION	Encryption where only the sender and receiver can decrypt messages, not the service in between.

STORING PASSWORDS SAFELY

- Good websites store the hash, not your password
- When you log in, they hash what you type and compare
- A salt (random extra data) is added so identical passwords get different hashes
- If a site can email you your password, it isn't storing it safely

END-TO-END ENCRYPTION

- Messages are encrypted on the sender's phone and only decrypted on the receiver's
- The app company's servers only see ciphertext
- Used by WhatsApp, Signal and iMessage
- Protects against hackers and snooping, even if the company is hacked

ACTIVITY · 20 MINUTES · PAIRS

HASH DETECTIVE

1. Use the online hash tool your teacher shows (or the printed table of hashes).
2. Hash the word "password", then "Password", then "password1". Compare the hashes.
3. Your teacher gives you a "stolen" list of unsalted hashes. Use the table of common password hashes to find which accounts used weak passwords.
4. Explain how salting would have protected them.

In your workbook, answer:

? What happened to the hash when you changed just one letter?

? Explain how a salt would stop the attack you just did.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. What does a hash function produce?

A An encrypted file with a key
B A longer password
C A picture
D A fixed-length fingerprint of the input

2. Why do websites store password hashes instead of passwords?

A Passwords are too long
B If stolen, hashes can't simply be turned back into passwords
C Hashes are shorter to type
D It's faster to log in

3. What is a salt?

A A type of virus
B A password manager
C Random data added to a password before hashing
D A spice for computers

4. With end-to-end encryption, who can read your messages?

A Only the sender and the receiver
B The government always
C Anyone on the Wi-Fi
D The app company

QUICK CHECK: ANSWERS

1. **D: A fixed-length fingerprint of the input**
Hashes are fixed-length fingerprints.

2. **B: If stolen, hashes can't simply be turned back into passwords**
Hashing protects stolen data.

3. **C: Random data added to a password before hashing**
Salts make identical passwords hash differently.

4. **A: Only the sender and the receiver**
Only the ends of the conversation.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how a website checks your password without storing it.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what a hash function does
- ✓ I can explain why passwords are hashed and salted
- ✓ I can describe end-to-end encryption

LESSON 6 OF 6

QUANTUM THREATS AND THE ENCRYPTION DEBATE

Powerful future quantum computers could break RSA and similar public-key systems, so the world is moving to new post-quantum cryptography.

STARTER · 5 MINUTES

DO NOW

Criminals copy encrypted messages today but can't read them. Why might they still want to keep them for 15 years?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain why quantum computers threaten today's public-key encryption and discuss the debate about encryption, privacy and security

SUCCESS CRITERIA: I CAN...

- ✓ I can explain the quantum threat to public-key encryption
- ✓ I can describe post-quantum cryptography
- ✓ I can argue both sides of the encryption backdoor debate

THE QUANTUM THREAT

- A large quantum computer could factor huge numbers quickly using Shor's algorithm
- That would break RSA and much of today's public-key encryption
- Today's quantum computers are far too small, but they are improving
- "Harvest now, decrypt later": attackers store encrypted data to read in future

KEY WORDS

QUANTUM COMPUTER	A computer that uses quantum physics to solve certain problems much faster than ordinary computers.
POST-QUANTUM CRYPTOGRAPHY	Encryption designed to stay secure even against quantum computers.
BACKDOOR	A secret way to get around security, such as a way for authorities to read encrypted messages.

POST-QUANTUM CRYPTOGRAPHY

- New encryption based on maths problems quantum computers can't easily solve
- In 2024 the US standards body NIST published the first post-quantum standards
- Governments, banks and tech companies are already switching
- Symmetric encryption like AES-256 is still considered strong

THE BACKDOOR DEBATE

- A backdoor: a secret way for authorities to read encrypted messages
- For: helps police catch criminals and protect children
- Against: a backdoor for the police is a weakness criminals could also use
- Should end-to-end encryption have exceptions? People strongly disagree

ACTIVITY · 25 MINUTES · WHOLE CLASS

ENCRYPTION DEBATE

1. Your teacher assigns roles: police officer, privacy campaigner, tech company boss, parent, journalist, cyber security expert.

2. Prepare two arguments for your character's view on "Messaging apps must let police read messages with a warrant".

3. Hold the debate (1 minute per role), then vote.

4. Write your own view afterwards.

In your workbook, answer:

? Write your character's two strongest arguments.

? Write your own view in 3-4 sentences, giving reasons and considering the other side.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. Why could quantum computers threaten RSA?

A They are faster at typing
B They can guess passwords
C They use more electricity
D They could factor huge numbers quickly

2. What is "harvest now, decrypt later"?

A Encrypting data twice
B Deleting old data
C Storing encrypted data today to decrypt with future computers
D Growing crops with computers

3. What is post-quantum cryptography?

A Encryption used after a computer is switched off
B A type of quantum computer
C Old encryption
D Encryption designed to resist quantum computers

4. What is an encryption backdoor?

A A door at the back of a data centre
B A strong password
C A secret way for authorities to read encrypted messages
D A type of firewall

QUICK CHECK: ANSWERS

1. **D: They could factor huge numbers quickly**
Shor's algorithm breaks factoring-based crypto.

2. **C: Storing encrypted data today to decrypt with future computers**
Long-term secrets are at risk.

3. **D: Encryption designed to resist quantum computers**
It's built on quantum-hard maths.

4. **C: A secret way for authorities to read encrypted messages**
Backdoors bypass encryption.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why "harvest now, decrypt later" is a concern for medical records.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain the quantum threat to public-key encryption
- ✓ I can describe post-quantum cryptography
- ✓ I can argue both sides of the encryption backdoor debate

END OF TOPIC

ENCRYPTION AND CRYPTOGRAPHY

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.